

Faith Frings

PHD CANDIDATE

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Education

Doctor of Philosophy, Integrative Biology

08/2022-present

UNIVERSITY OF NEW HAMPSHIRE

- Advisor: Dr. Melissa Aikens | Dissertation: Building data literate biologists: Investigating student motivational dynamics in statistical programming across diverse biology courses

Master of Science, Biology with a concentration in ecology and evolution

08/2020-08/2022

UNIVERSITY OF SOUTH FLORIDA

- Advisor: Dr. Luanna Prevost | Thesis: Mixed-method analysis of undergraduate student interpretation of different phylogenetic tree styles

Bachelor of Science, Marine Biology

08/2017-05/2020

UNIVERSITY OF SOUTH FLORIDA

Research Experience

DISSERTATION (2022 - PRESENT)

Advised by Dr. Melissa Aikens

CHARACTERIZING STUDENT R PROGRAMMING MOTIVATION AND ENGAGEMENT

- Conducted multilevel modeling to examine changes in students' motivation to learn R programming
- Conducted path analysis to investigate relationships among students' R programming mindsets, task values, and effort on R programming tasks

EXAMINING STUDENT R PROGRAMMING MOTIVATION ACROSS COURSE TYPES

- Collected and analyzed open-response survey data to identify factors shaping students' motivation across R-integrated biology courses

EXPLORING STUDENT MOTIVATION TO LEARN DATA SCIENCE IN THE FIELD

- Collected and analyzed semi-structured interviews to identify factors shaping students' motivation to learn R programming in a field-based data science course

FACTORS EXPLAINING STUDENT R PROGRAMMING RESOURCE USE IN BIOLOGICAL RESEARCH

- Collected and analyzed semi-structured interviews, participant observations, and surveys to examine how and why students select resources when programming in a biological research course

THESIS (2020 - 2022)

Advised by Dr. Luanna Prevost

EXAMINING STUDENT PHYLOGENETIC TREE INTERPRETATIONS ACROSS TREE STYLES

- Conducted thematic analysis, phi correlations, and Mann-Whitney U tests of open-response data to characterize undergraduate students' reasoning across different phylogenetic tree styles

UNDERGRADUATE RESEARCH (2019 - 2020)

Advised by Dr. Luanna Prevost

STUDENT REASONING ABOUT FOOD WEBS

- Conducted thematic analysis and phi correlations of open-response data examining students' interpretations of food webs

GRADUATE RESEARCH ASSISTANT

Biology Education Research Group (BERG), UNH

COMMUNITY COLLEGE TRANSFER STUDENT PERSISTENCE

2022-2026

- Conducted semi-structured interviews and thematic analysis to investigate supports and barriers experienced by community college transfer students at a four-year institution

Biology Education Research Group (BERG), UNH

STUDENT GRAPHICAL REASONING

2022-2026

- Conducted thematic analysis of open-response data to characterize students' graphical reasoning

Prevost Lab, USF

AMGEN BIOTECHNOLOGY EDUCATION PROJECT

2022

- Developed online instructional resources for biology educators through an Amgen Foundation-funded biotechnology education project

Teaching Experience

PRIMARY INSTRUCTOR

BIOSM 2350/MEFB 633: R by the Sea: Data Analysis in Marine Sciences

06/2026

UNDERGRADUATE, 1 SUMMER TERM

GRADUATE TEACHING ASSISTANT

BIOL 706/806: Data Science for Life Sciences

2023 - 2026

UNDERGRADUATE/GRADUATE, 4 SEMESTERS

BIOL 528: Applied Biostatistics

2023 - 2026

UNDERGRADUATE, 4 SEMESTERS

BIOSM 3290/MEFB 714: Field Animal Behavior

2025

UNDERGRADUATE, 1 SUMMER TERM

BIOSM 4990/MEFB 751: Research in Biology

2024

UNDERGRADUATE, 1 SUMMER TERM

BIOSM 1610/MEFB 674: Ecology & the Marine Environment

2024

UNDERGRADUATE, 1 SUMMER TERM

BIOL 412: Introductory Biology Laboratory

2022

UNDERGRADUATE, 1 SEMESTER

BSC 2011: Biology II – Biological Diversity

2022

UNDERGRADUATE, 1 SEMESTER

BSC 2011L: Biology II – Biological Diversity Laboratory

2020 - 2021

UNDERGRADUATE, 2 SEMESTERS

BSC 1005: Biological Principles for Non-Majors

2021

UNDERGRADUATE, 1 SEMESTER

WORKSHOPS

R by the Sea - Japan. Advanced Institute for Marine Ecosystem Change (WPI-AIMEC), Tohoku University.

01/2026

CO-TAUGHT A FIVE-DAY INTRODUCTORY R AND STATISTICS WORKSHOP

INVITED LECTURES

3. **Frings, F.**, (2026, September). *Introduction to different styles and applications of phylogenetic trees*. Introductory Biology: Evolution, Biodiversity and Ecology (BIOL 412). University of New Hampshire.
2. **Frings, F.**, & Caldwell, A. (2026, September). *Graduate school advice & individual development plans*. Introductory Graduate Seminar (BIOL 901). University of New Hampshire.
1. **Frings, F.**, & Villeneuve, D. (2025, September). *Graduate school advice & individual development plans*. Introductory Graduate Seminar (BIOL 901). University of New Hampshire.

INVITED FACILITATOR

Professional Development in Evidence-Based Teaching Series (PDET)

2026

Teaching Distinctions & Professional Development

UNH Graduate Student Excellence in Teaching Award

2025

RECOGNITION BY THE DEPARTMENT OF BIOLOGICAL SCIENCES FOR TEACHING ACHIEVEMENT

Human Subjects Research Certificate

2019,22,25

WEB-BASED COURSE MODULES ON RESPONSIBLE CONDUCT OF HUMAN SUBJECTS RESEARCH

Journal Articles

2. **Frings, F.**, Heim, A., & Prevost, L. (in preparation). *Mixed-method analysis of undergraduate student interpretation of different phylogenetic tree styles.*
1. **Frings, F.**, Weigel, E. G., & Aikens, M. L. (in review). *Preparing data-literate biologists: Examining motivational factors that drive student effort in R programming.*

Book Chapters

1. Esparza, D., Amin, N. E., **Frings, F.**, Halmo, S. M., Hazlett, Z. S., Heim, A. B., Uminski, C., Von der Mehden, B., Wright, A. M., & Yang, M. (accepted). *Guiding discipline-based education research trainees through career transitions with inclusive and accessible organizational support* (C. Etson, T. Lightner, & R. Greenler, Eds.; 1st ed.). Lexington Books.

Media

1. **Frings, F.**, Levine, A., Madrigal, L., Plank, L., & Richards, C. (2021). *Celebrating the importance of evolution.* Frontiers for Young Minds.

Presentations

Underlined names indicate undergraduate research mentees.

15. **Frings, F.**, Weigel, E. G., & Aikens, M. L. (2026, July). *Preparing data literate biologists: What drives student effort in R programming?* [Talk]. The Society for the Advancement of Biology Education Research 16th Annual Meeting. East Lansing, MI, USA.
14. Koopman, C., Greene, A., **Frings, F.**, & Aikens, M. L. (2026, April). *Motivating factors for students learning R programming in STEM courses.* [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.
13. **Frings, F.**, Weigel, E. G., & Aikens, M. L. (2026, February). *Preparing data literate biologists: What drives student effort in R programming?* [Talk]. University of New Hampshire, Department of Biological Sciences Spring Seminar Series.
12. **Frings, F.**, & Aikens, M. L. (2025, July). *Exploring student motivation in programming: Insights from a hybrid field-based data science course.* [Poster]. The Society for the Advancement of Biology Education Research 15th Annual Meeting. Minneapolis, MN, USA.
11. Dearth, A., **Frings, F.**, & Aikens, M. L. (2025, April). *Investigating student engagement in statistical programming in biology.* [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.
10. **Frings, F.**, & Aikens, M. L. (2025, April). *Characterizing the biology students' mindsets and task values towards programming.* [Talk]. The University of New Hampshire Annual Graduate Research Conference.
9. Maddox, M., Prieto, A., **Frings, F.**, & Aikens, M. L. (2025, April). *Exploring students' programming ability beliefs in a data science course.* [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.
8. **Frings, F.**, & Aikens, M. L. (2025, February). *Characterizing the biology students' mindsets and task values towards programming.* [Talk]. University of New Hampshire, Department of Biological Sciences' Spring Seminar Series.
7. **Frings, F.**, & Aikens, M. L. (2024, July). *Community college transfer students' perceptions of supports and barriers to persistence in their major at a four-year institution.* [Poster]. The Society for the Advancement of Biology Education Research 14th Annual Meeting. Minneapolis, MN, USA.

6. Amovic, S., Maddox, M., **Frings, F.**, Schuchardt, A., & Aikens, M. L. (2024, April). *Analysis of biology students’ cognitive challenges when learning statistics in a course-based undergraduate research experience*. [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.
5. Weiss, V., **Frings, F.**, & Aikens, M. L. (2023, April). *Supports and barriers affecting community college transfer student degree persistence at a four-year institution*. [Poster]. The University of New Hampshire Annual Undergraduate Research Conference.
4. **Frings, F.**, & Prevost, L. P. (2022, October). *Mixed method analysis of undergraduate student interpretation of different phylogenetic tree styles*. [Talk]. University of New Hampshire, Department of Biological Sciences Fall Seminar Series.
3. **Frings, F.**, & Prevost, L. P. (2022, March). *Mixed method analysis of undergraduate student interpretation of different phylogenetic tree styles*. [Poster]. The 4th Biennial CIRCLE Research Conference. St. Louis, MO, USA.
2. **Frings, F.**, & Prevost, L. P. (2021, March). *Mixed method analysis of undergraduate student interpretation of different phylogenetic tree styles*. [Poster]. The 113th Sigma Xi Annual Meeting & Student Research Conference. Virtual Conference.
1. **Frings, F.**, Grissett, C., & Prevost, L. P. (2020, April). *Correlation between competition and arrow reversals in students’ responses to conceptual food web questions*. [Poster]. The University of South Florida 18th Annual Undergraduate Research Conference. Virtual Conference.

Research Mentoring

UNDERGRADUATE MENTEES

ASHLEY GREENE	2026
CARLEIGH KOOPMAN	2025-2026
ABIGAIL DEARTH	2025
AVA PRIETO	2025
SARA AMOVIC	2023-2024
MEILING MADDOX	2023-2025
VIRGINIA WEISS	2022 - 2023

Awards & Fellowships

Student Teaching Award Fellowship	2025
UNH GRADUATE SCHOOL, UNIVERSITY OF NEW HAMPSHIRE	\$2000
Bill Woods Graduate Student Poster Award	2024
SOCIETY FOR THE ADVANCMENT OF BIOLOGY EDUCATION RESEARCH ANNUAL MEETING	\$500
Travel Award	2024, 2025
DEPARTMENT OF BIOLOGICAL SCIENCES, UNIVERSITY OF NEW HAMPSHIRE	\$800
Travel Award	2024, 2026
UNH GRADUATE SCHOOL, UNIVERSITY OF NEW HAMPSHIRE	\$800
S-STEM Scholarship	2020-2021
INTEGRATIVE BIOLOGY DEPARTMENT, UNIVERSITY OF SOUTH FLORIDA	\$5,600

Grants in Support of Research

Graduate Student Research Fund	2025
SCHOOL OF MARINE SCIENCES AND OCEAN ENGINNERING, UNIVERSITY OF NEW HAMPSHIRE	\$998

Professional Service

UNH R USERS GROUP ORGANIZER

- Organize and lead weekly R Users Group meetings

2023-present

SABER VIRTUAL DAY CONFERENCE COMMITTEE

- Co-developed and co-hosted the inaugural Virtual Day of the 16th SABER annual meeting

2025 - 2026

SABER DISCIPLINE-BASED EDUCATION RESEARCH-SCHOLARS IN TRAINING (DBER-SiT) CHAIR

- Organized and delegated DBER-SiT committee projects

2025-2026

SABER DBER-SiT CHAIR-ELECT

- Shadowed and assisted the Chair in their role

2024-2025

SABER DBER-SiT FUNDING CO-CHAIR

- Identified and disseminated funding sources for DBER-scholars in training

2023-2025

UNH UNDERGRADUATE RESEARCH CONFERENCE POSTER JUDGE

- Evaluated quality of research and design of undergraduate life sciences research posters

2023-2026

UNH DBS GRADUATE STUDENT COMMITTEE

- Co-founded the committee; organized graduate student seminar series; hosted outreach events

2022-2024

Manuscripts Reviewed

Tips, Tricks, and Techniques in the American Biology Teacher

Technical Skills

- **Programming:** R (advanced)
- **Quantitative methods:** Multilevel modeling, structural equation modeling
- **Qualitative methods:** Thematic analysis, cross-case analysis, semi-structured interview protocol development, participant observation, open-response item development
- **Reproducible research:** R Markdown, Quarto
- **Version control:** Git/GitHub

Professional Society Memberships

Society for the Advancement of Biology Education Research

References

Melissa Aikens, Ph.D.

DEPARTMENT ASSOCIATE CHAIR, ASSOCIATE PROFESSOR OF BIOLOGY EDUCATION; UNIVERSITY OF NEW HAMPSHIRE

Melissa.Aikens@unh.edu

Easton White, Ph.D.

ASSOCIATE PROFESSOR, DEPARTMENT OF BIOLOGICAL SCIENCES, ASSOCIATE DIRECTOR OF RESEARCH, SCHOOL OF MARINE SCIENCE AND OCEAN ENGINEERING, UNIVERSITY OF NEW HAMPSHIRE

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Luanna Prevost, Ph.D.

TAMPA CAMPUS ASSOCIATE CHAIR, ASSOCIATE PROFESSOR, DEPARTMENT OF INTEGRATIVE BIOLOGY, UNIVERSITY OF SOUTH FLORIDA

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